Restaurant Recommendation System Using Machine Learning

Machine Learning Internship Project Report

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1. Introduction

With thousands of restaurants available across different cities, selecting a restaurant that matches a user's preferences can be challenging. Recommendation systems help users discover restaurants that best suit their interests by analyzing available restaurant information and identifying similar options.

This project develops a **Content-Based Restaurant Recommendation System** using machine learning techniques. The system recommends restaurants based on features such as cuisine, city, and price range. It applies text vectorization and similarity measurement to identify restaurants with similar characteristics and provide personalized recommendations.

2. Objective

The objectives of this project are:

- Develop a content-based restaurant recommendation system.
- Perform data preprocessing and cleaning.
- Explore and analyze restaurant data through visualization.
- Generate recommendations using restaurant features.
- Measure similarity between restaurants using machine learning techniques.
- Provide accurate and relevant restaurant suggestions based on user preferences.

3. Dataset Description

Dataset Information

Attribute	Value
Dataset Name	Zomato Restaurant Dataset
Total Records	9551
Total Features	21
Recommendation Type	Content-Based Filtering

Important Features Used

- Restaurant Name
- City
- Cuisines
- Average Cost for Two
- Price Range
- Aggregate Rating
- Votes
- Has Online Delivery
- Has Table Booking

The dataset contains restaurant information collected from multiple cities and includes details related to cuisine, pricing, customer ratings, and restaurant services.

Restaurant Name	City	Cuisines	Aggregate rating	Price range
Le Petit Souffle	Makati City	French, Japanese, Desserts	4.8	3
Izakaya Kikufuji	Makati City	Japanese	4.5	3
Heat - Edsa Shangri-La	Mandaluyong City	Seafood, Asian, Filipino, Indian	4.4	4
Ooma	Mandaluyong City	Japanese, Sushi	4.9	4
Sambo Kojin	Mandaluyong City	Japanese, Korean	4.8	4

4. Data Preprocessing

The following preprocessing steps were performed before building the recommendation system:

- Loaded the dataset using Pandas.
- Checked for missing values.
- Replaced missing values in the **Cuisines** column with **"Unknown"**.
- Removed duplicate records, if any.
- Selected relevant features for recommendation.
- Combined selected features into a single text feature.
- Converted text data into numerical vectors using **CountVectorizer**.

These preprocessing steps prepared the dataset for similarity analysis.

5. Exploratory Data Analysis (EDA)

Several visualizations were created to better understand the dataset.

Visualizations Included

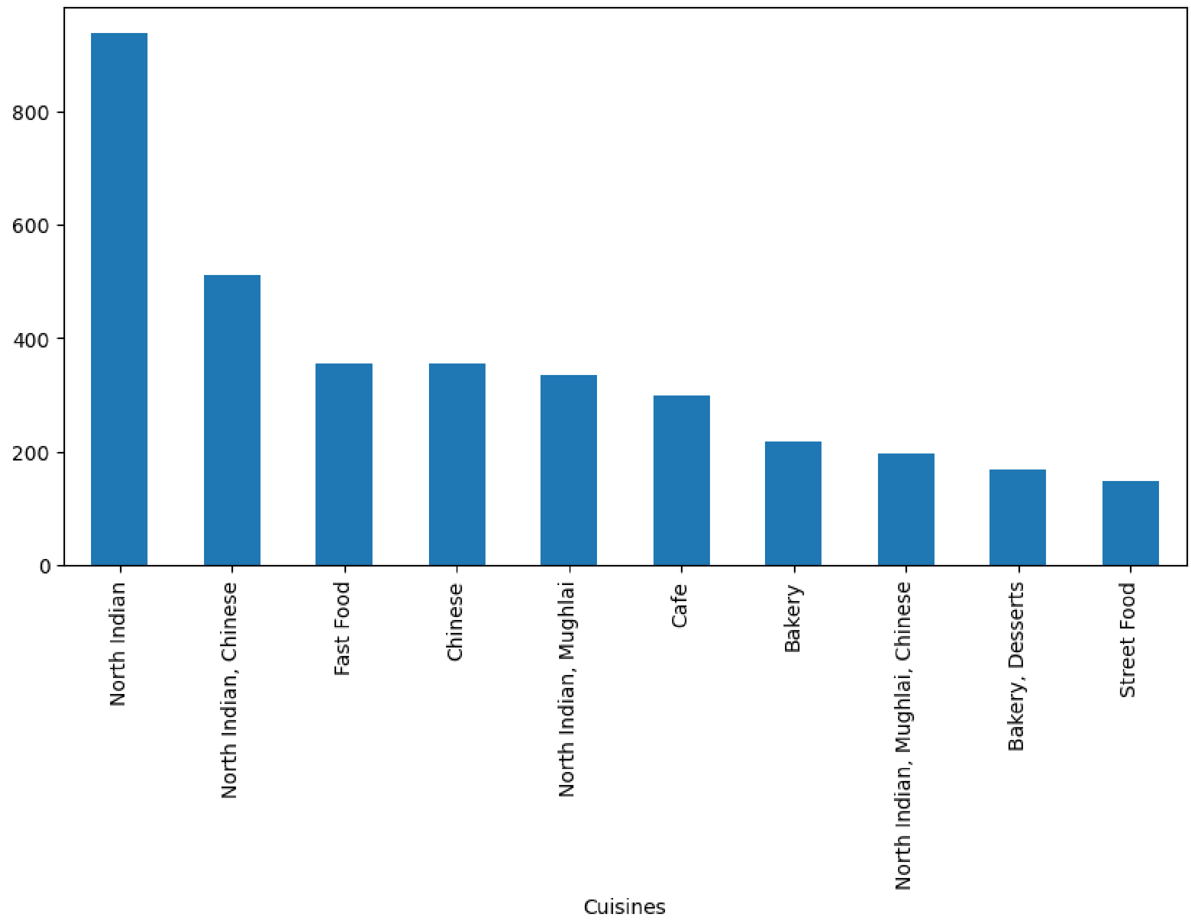
- Rating Distribution
- Top Cities by Number of Restaurants
- Most Popular Cuisines
- Price Range Distribution
- Votes vs Aggregate Rating
- Correlation Heatmap

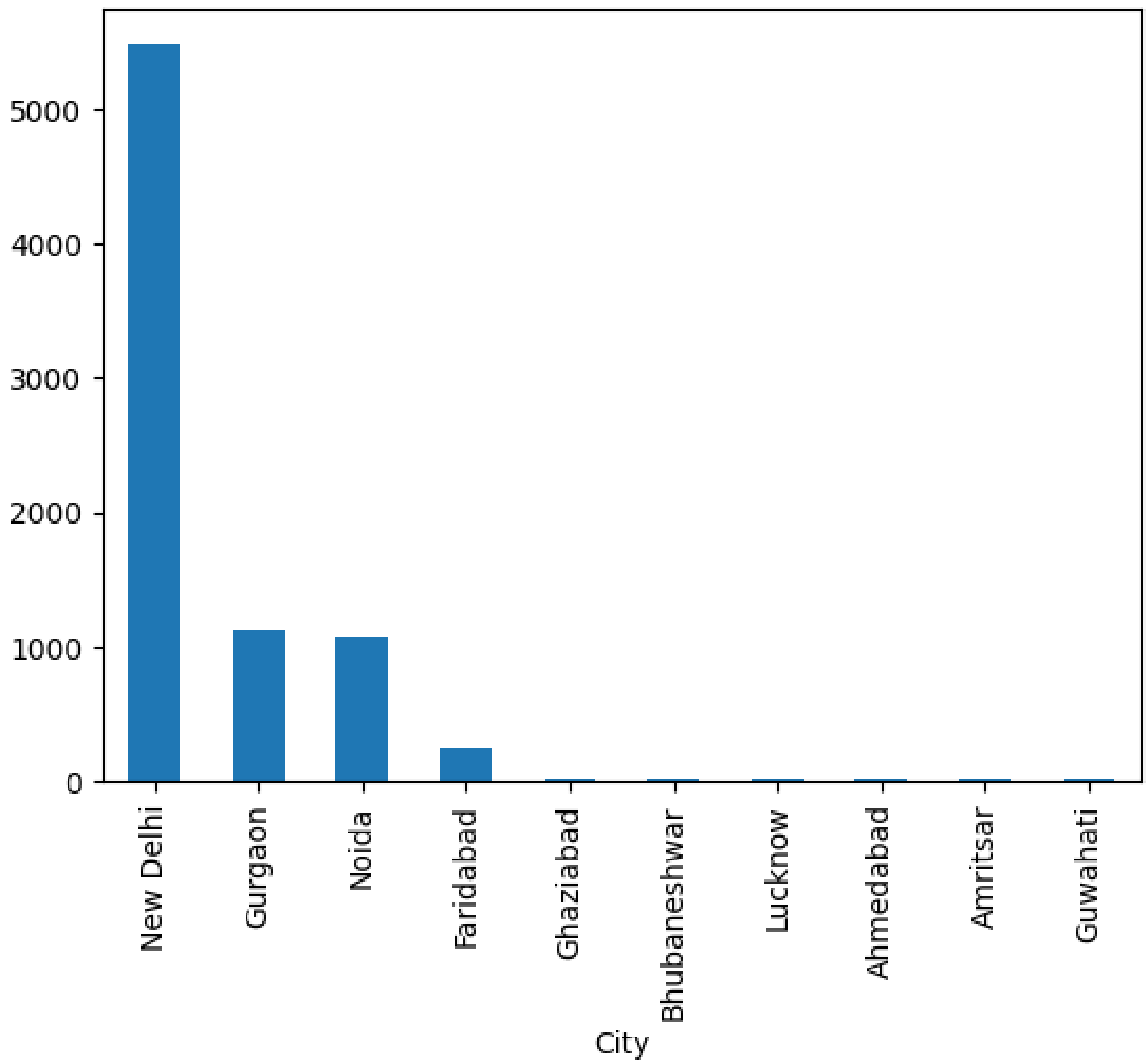
Observations

- Most restaurants have ratings between **3.5 and 4.5**.
- North Indian and Chinese cuisines are the most common.
- Restaurants with more customer votes generally have higher ratings.
- Medium-priced restaurants appear more frequently in the dataset.

- Online delivery is available for many highly rated restaurants.

	Restaurant Name	City	Cuisines	Price range	Aggregate rating	Combined_Features	Similarity
0	Le Petit Souffle	Makati City	French, Japanese, Desserts	3	4.8	Makati City French, Japanese, Desserts 3	0.0
1	Izakaya Kikufuji	Makati City	Japanese	3	4.5	Makati City Japanese 3	0.0
2	Heat - Edsa Shangri-La	Mandaluyong City	Seafood, Asian, Filipino, Indian	4	4.4	Mandaluyong City Seafood, Asian, Filipino, Ind...	0.0
3	Ooma	Mandaluyong City	Japanese, Sushi	4	4.9	Mandaluyong City Japanese, Sushi 4	0.0
4	Sambo Kojin	Mandaluyong City	Japanese, Korean	4	4.8	Mandaluyong City Japanese, Korean 4	0.0





6. Content-Based Recommendation System

This project implements a **Content-Based Filtering** recommendation approach.

Methodology

1. Selected important restaurant features.
2. Combined the selected features into a single text representation.
3. Converted text into numerical vectors using **CountVectorizer**.
4. Calculated similarity scores using **Cosine Similarity**.
5. Recommended restaurants with the highest similarity scores.

Workflow

Restaurant Dataset

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Data Cleaning

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Feature Selection

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Combined Features

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CountVectorizer

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Cosine Similarity

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Top Restaurant Recommendations

7. Recommendation Results

The recommendation system successfully suggests restaurants with similar characteristics.

Example User Preference

Feature	Value
City	New Delhi
Cuisine	North Indian
Price Range	3

Sample Recommendations

- Kopper Kadai
- Zabardast Indian Kitchen

- Band Baaja Baaraat
- Pind Balluchi
- The GT Road

The recommended restaurants closely matched the selected cuisine, city, and pricing preferences.

	Restaurant Name	City	Cuisines	Price range	Aggregate rating	Combined_Features	Similarity
6656	Kopper Kadai	New Delhi	North Indian	3	4.8	New Delhi North Indian 3	1.0
3014	Zabardast Indian Kitchen	New Delhi	North Indian	3	4.7	New Delhi North Indian 3	1.0
6655	Band Baaja Baaraat	New Delhi	North Indian	3	4.6	New Delhi North Indian 3	1.0
3999	Rang De Basanti Urban Dhaba	New Delhi	North Indian	3	4.4	New Delhi North Indian 3	1.0
4087	Bukhara - ITC Maurya	New Delhi	North Indian	4	4.4	New Delhi North Indian 4	1.0

8. Evaluation

The recommendation system was evaluated by testing different combinations of user preferences.

Evaluation Criteria

- Similarity of recommended restaurants
- Relevance to selected cuisine
- Matching city preferences
- Price range consistency

Findings

- Recommendations were highly relevant for common cuisines.
- Restaurants recommended belonged to similar categories and locations.
- Content-Based Filtering effectively identified restaurants with comparable characteristics.
- The system generated meaningful recommendations for different user preferences.

9. Conclusion

This project successfully developed a **Content-Based Restaurant Recommendation System** using machine learning techniques. By applying CountVectorizer and Cosine Similarity, the system generated personalized restaurant recommendations based on cuisine, city, and price range.

The recommendation engine demonstrated the effectiveness of content-based filtering in identifying restaurants with similar characteristics and can serve as a foundation for more advanced recommendation systems.

10. Future Scope

The recommendation system can be enhanced through the following improvements:

- Hybrid Recommendation System
- Collaborative Filtering
- TF-IDF Vectorization
- Deep Learning-Based Recommendation Models
- Personalized User Profiles
- Real-Time Restaurant Recommendation
- Streamlit Web Application Deployment
- Integration with Restaurant APIs

11. References

1. Cognifyz Technologies – Machine Learning Internship Guidelines.
2. Scikit-learn Documentation – CountVectorizer and Cosine Similarity.